

## PCD| Homework 1

Write a program that measures the time required to transfer various amounts of data (500 MB, 1 GB) using different transport protocols and different transfer methods. The program should include detailed measurements to analyze the performance of distributed networks using **TCP**, **UDP**, and **QUIC** protocols.

Implementation Requirements:

### 1. Supported Protocols (as parameters for client and server):

- **TCP** (Transport Control Protocol):
  - Provides reliable connections with flow control and in-order packet delivery.
- **UDP** (User Datagram Protocol):
  - Provides fast transport without flow control or error handling mechanisms.
- **QUIC** (Quick UDP Internet Connections):
  - Provides fast, reliable, and secure connections with built-in multiplexing, avoiding head-of-line blocking issues.
  - You can use one of the following libraries: **ngtcp2**, **quiche**, **lsquic**, et.al.

### 2. Message Size:

- Transfer data in blocks with sizes ranging from **1** to **65535 bytes**.
- Test different message sizes within this range for each protocol (TCP, UDP, and QUIC).

### 3. Data Transfer Amounts:

- Transfer **500 MB** and **1 GB** of data.
- You may use a large reusable buffer or simulate real-world data structures (e.g., "1 GB = 1.5 million WhatsApp messages, 4000 photos, or 10,000 emails").

### 4. Transfer Mechanisms:

- **Streaming:**
  - Continuous transfer of data without waiting for confirmation after each message.
- **Stop-and-Wait:**
  - Acknowledgment for each message before sending the next one.
  - Implementing **stop-and-wait** for **TCP** is optional since TCP already manages acknowledgments and retransmissions automatically.
  - Implementing **stop-and-wait** for **UDP** and **QUIC** is mandatory to analyze its impact on performance and data loss.

Output Requirements:

### 1. The Server must display the following at the end of each session:

- The protocol used: **TCP**, **UDP**, or **QUIC**.
- The total number of messages received.

- The total number of bytes received.
- 2. **The Client** must display the following at the end of the execution:
  - Total transmission time (in seconds).
  - The total number of messages sent.
  - The total number of bytes sent.
- 3. **Testing and Report:**
  - Students must perform comparative tests using all three protocols (TCP, UDP, QUIC) for various message sizes and data amounts (500 MB and 1 GB).
  - The analysis must include detailed comparisons of:
    - Transmission time for each protocol.
    - Data loss rates (for UDP and QUIC).
    - The impact of message size on performance.

#### Evaluation Criteria:

1. Complete, functional client and server code that runs on a **Linux** system.
2. A PDF/Word/Text document containing:
  - Detailed information on how to use the program (options for client/server).
  - Statistics and comparisons of performance for each protocol.
  - A detailed analysis of the experimental results.
  - Conclusions about the advantages and disadvantages of each protocol under different transfer conditions.

#### General Observations:

1. If the buffer size for **UDP** is increased, data loss may be minimal in a **LAN**, but packet loss may occur in more complex networks.
2. The **stop-and-wait** mechanism is not very useful for **TCP**, as TCP already handles acknowledgments, but it may be interesting to analyze its impact.
3. The **stop-and-wait** mechanism is more relevant for **UDP** and **QUIC**, especially in contexts where message loss needs to be minimized.

## Instrucțiuni de trimitere a temei

### Format arhivă:

- **Nume fișier:** Tema1\_NumePrenume\_2025\_AcronimMaster.zip (fără diacritice)
- **Exemplu:** Tema1\_PopescuIon\_2025\_MSD.zip
- **Conținut:**
  - Cod sursă - propriu (fără librării)
  - Documentul PDF/Word/Text
- **Locație:** URL-ul indicat în formularul de înregistrare unde puteți să modificați link-ul trimis și **DOAR** dacă este absolut necesar  
(<https://docs.google.com/forms/d/e/1FAIpQLScvC2VxwqsXr02TXlwQjIT7-NL6YgJLHq6P5wuonhb265C8MQ/viewform>)

